Week 2: The Paper's Simple Transformer

Reading Project: Near-Optimal Real-Time Personalization with Simple Transformers

Goal for the week

Implement the paper's simple transformer and explain its two averaging properties. Show how the selected set, user vector, and activation functions determine the output.

Budget: 6–8 hours, including roughly one hour writing.

1 Reading

Read **Section 2.2, including Definition 2** (pp. 13–15) of *Near-Optimal Real-Time Personalization with Simple Transformers*, v2. https://arxiv.org/html/2503.00608v2

David presents Section 2.2

2 The object we are building

For a nonempty set $S \subseteq \{1, \dots, n\}$, let $\boldsymbol{X}_S = \operatorname{diag}(\mathbf{1}_{i \in S})$. As in Week 1, $\boldsymbol{Q}, \boldsymbol{K} \in \mathbb{R}^{n \times d_{kq}}$ and $\boldsymbol{V} \in \mathbb{R}^{n \times d_v}$. Now add $u \in \mathbb{R}^{d_v}$ and non-decreasing functions $f_i : \mathbb{R} \to \mathbb{R}$.

Masking clarification. The printed formula zeroes unselected rows but leaves their columns in softmax. This disagrees with its averaging explanation. Here we explicitly exclude unselected columns before softmax, matching that explanation and the selected-item formula in Observation 2. Define

$$L = (\boldsymbol{X}_S \boldsymbol{Q})(\boldsymbol{X}_S \boldsymbol{K})^\top, \qquad (M_S)_{ij} = \begin{cases} 0, & j \in S, \\ -\infty, & j \notin S, \end{cases}$$

$$B_S = \operatorname{softmax}(L + M_S), \qquad A_S = B_S \boldsymbol{X}_S \boldsymbol{V}, \qquad T_i = f_i\big((A_S u)_i\big).$$

Softmax is row-wise, with $\exp(-\infty) = 0$. Keep zeroed query rows for $i \notin S$ and the paper's unscaled dot products.

Example for Q3 and Q4

Use $n = 3$, $d_{kq} = d_v = 1$, and

$$\boldsymbol{Q} = \begin{pmatrix} \log 2 \\ 0 \\ 1 \end{pmatrix}, \quad \boldsymbol{K} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad \boldsymbol{V} = \begin{pmatrix} 2 \\ 8 \\ 20 \end{pmatrix}, \quad u = 2, \quad f_i(t) = \max\{0, t - 3\}.$$

Start with $S = \{1, 2\}$. Here log is the natural logarithm.

3 Questions

Answer in your own words. Keep all written answers to **250 words total**, plus equations and the table. Code and full test output belong in the repository.

Q1. From a set to user-specific rewards

From memory, explain $S \to \boldsymbol{X}_S \to A_S \to T$. Give the shapes of A_S and T, and the roles of u and f_i. Does changing only u change attention? Check your explanation and note any correction. *Up to 60 words.*

S is the chosen set of items. $\boldsymbol{X}_S$ zeroes the rows of unchosen items. Attention gives A_S, which is $n \times d_v$. Multiplying by u gives one number per item, and f_i turns it into a reward, so T has length n. Changing only u doesn't change attention. After reviewing my initial answer, my correction was that the unselected columns must be masked and not just zeroed.

Q2. The two averaging properties

Derive $(A_S)_i$ for $i \in S$ and $i \notin S$ under the stated mask. Why is the first a convex combination? Explain the uniform average in two sentences. *Two equations and up to 60 words.*

$$(A_S)_i = \sum_{j \in S} \frac{\exp(q_i^\top k_j)}{\sum_{l \in S} \exp(q_i^\top k_l)} v_j, \qquad i \in S,$$

$$(A_S)_i = \frac{1}{|S|} \sum_{j \in S} v_j, \qquad i \notin S.$$

For $i \in S$, softmax runs over only S. The weights are positive and sum to 1, so the output is a weighted average (a convex combination) of selected values. For $i \notin S$, the query row is zero, so all scores are 0, the weights are equal, and the output is the average of the selected values.

Q3. What changes when the set changes?

Predict, calculate by hand, then verify with code. Why can adding item 3 change item 1's reward? *Table and up to 40 words; note any hand/code discrepancy.*

Set	Row	Weights on $(1, 2, 3)$	$(A_S)_i$	T_i
$\{1, 2\}$	$i = 1$	(2/3, 1/3, 0)	4	5
$\{1, 2\}$	$i = 3$	(1/2, 1/2, 0)	5	7
$\{1, 2, 3\}$	$i = 1$	(4/7, 2/7, 1/7)	$44/7 \approx 6.29$	$67/7 \approx 9.57$

Adding item 3 adds a column to row 1's softmax. Its value is big (20), so row 1's average goes up: A_1 from 4 to 6.29, T_1 from 5 to 9.57. My calculations by hand and through the code verification all matched.

Q4. Why a separate value matrix?

Return to $S = \{1, 2\}$. Predict the effect of $\boldsymbol{V} \to 2\boldsymbol{V}$ on B_S, A_S, and T; check row 1. Does T double? Explain how $\boldsymbol{V}$'s role differs from $\boldsymbol{Q}$ and $\boldsymbol{K}$. *Up to 40 words.*

B_S stays the same (it only depends on $\boldsymbol{Q}$ and $\boldsymbol{K}$). A_S doubles. T doesn't: T_1 goes 5 to 13, since f subtracts 3 after 8 becomes 16. $\boldsymbol{Q}$ and $\boldsymbol{K}$ decide where to look; $\boldsymbol{V}$ is what gets averaged.

4 Code

Reuse Week 1's repository and stable softmax; keep its unmasked function for comparison.

Task 1: `simple_transformer.py`

Implement $\boldsymbol{X}_S$, the mask, and the full map through u and item-specific f_i. Return B_S, A_S, and T.

Task 2: verify the two facts

In `tests/test_simple_transformer.py`, use several random instances with $d_{kq}, d_v > 1$ and check:

1. Check nonnegative weights, unit row sums, zero unselected-column weights, and the selected-row weighted-sum formula. Coordinate bounds alone are insufficient.
2. Every unselected output row equals the mean of the selected value vectors.
3. A singleton $S = \{i\}$ returns v_i before activation. Reject an empty S for this week's attention function.
4. The full map works with identity and non-decreasing nonlinear activations, including different f_i for different items.

Task 3: verify the example

Verify Q3 and Q4. Compare an unselected row with Week 1's unmasked function; explain in a code comment why zero values still affect its softmax denominator.

5 Write-up

Fill Q1–Q4 and the fields below. Commit the `.tex` file and code; send the PDF **24 hours before the meeting**. Only answers count toward 250 words.

W1. One question for our meeting

One specific sentence, or "none after checking."

I had a question about Section 2.2 in the main paper. In Section 2.2, unselected rows are zeroed by X_S, but their columns stay in the softmax with logit 0. Doesn't that break the claim that the weights are uniform over only S?

W2. Evidence and time

Repository/commit link, command to run the checks, and total hours. One line.

https://github.com/dwc2425/transformer-personalization; run `python -m pytest`; this took about 4 hours total.

Meeting preparation

Prepare a ten-minute explanation without notes: derive both averaging properties, walk through the full model, and explain your set-change example.